# Data Analysis Report

## Financial Loan

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06/March/2026

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# COMPANY INTRODUCTION

**Apex Lending Group** is a fast-growing consumer lending company that provides personal loans, debt consolidation loans, and small business loans across the United States. In 2021 we issued 38,576 loans worth millions, but our default and late-payment rates are higher than industry benchmarks. Our risk team is under pressure to improve loan approval decisions, pricing strategy, and collection processes before the next funding round.

# PROBLEM

- Too many loans are becoming delinquent or defaulting, especially in certain grades, purposes, and income segments.
- We want to understand which customer profiles (income, employment length, home ownership, purpose) are safest vs riskiest.
- Interest rate (int_rate) and installment amount seem to influence repayment behavior.
- We need clear insights on **Debt-to-Income (DTI)** impact, loan purpose risk, and grade vs actual performance.
- **Goal**: Reduce bad loans by 15–20% in 2026 while maintaining healthy loan volume.

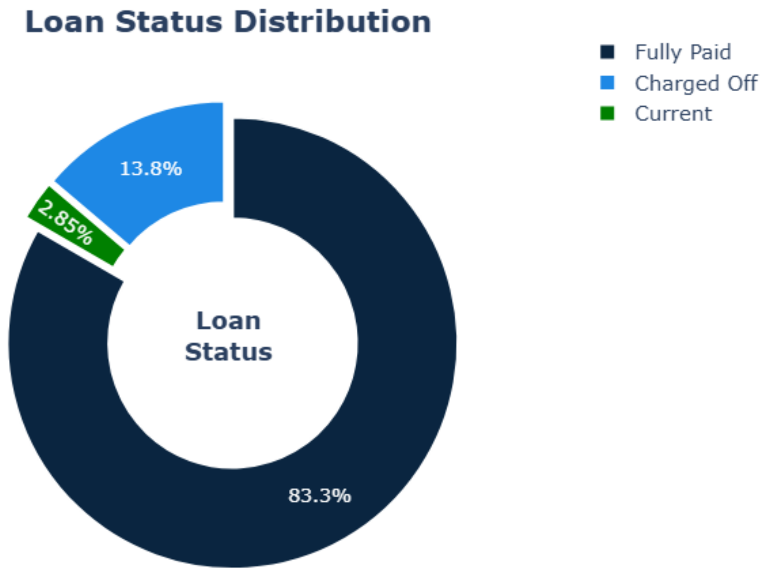
# KEY BUSINESS QUESTIONS

1. Overall performance
2. Customer & Risk Segmentation
3. Pricing & Repayment Behavior
4. Geographic & Behavioral Insights
5. Revenue & Risk Signals
6. Business Recommendations(Suggestions)

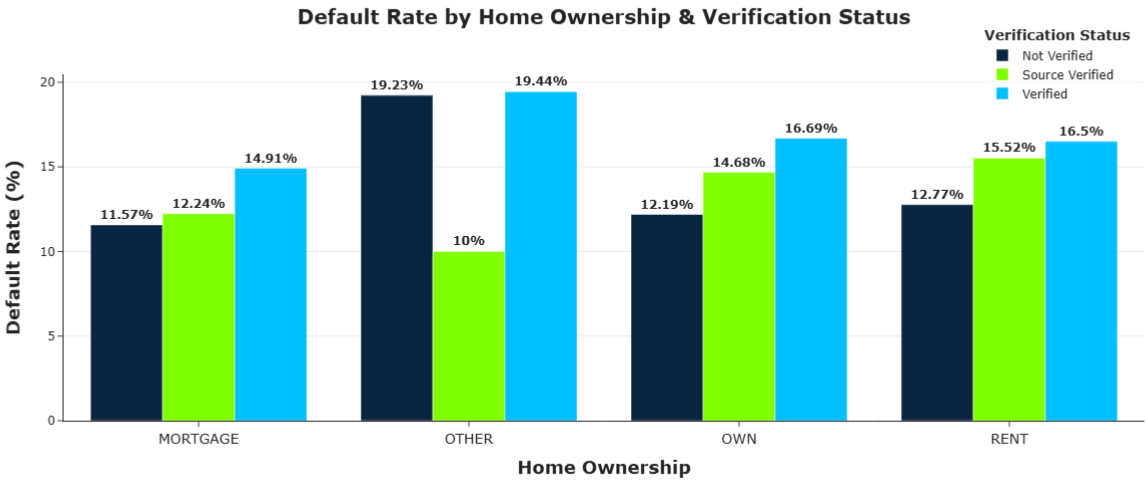
# KEY FINDINGS

## OVERALL PERFORMANCE

| | | | |
|---|---|---|---|
| Total Loans Issued | 38,576 | Total Amount Lent | $435,757,075 |
| Average Loan Size | $11,296.07 | Average Interest Rate | 12.05% |
| Total Payment Received | $473,070,933 | Overall Default Rate | 13.82 % |

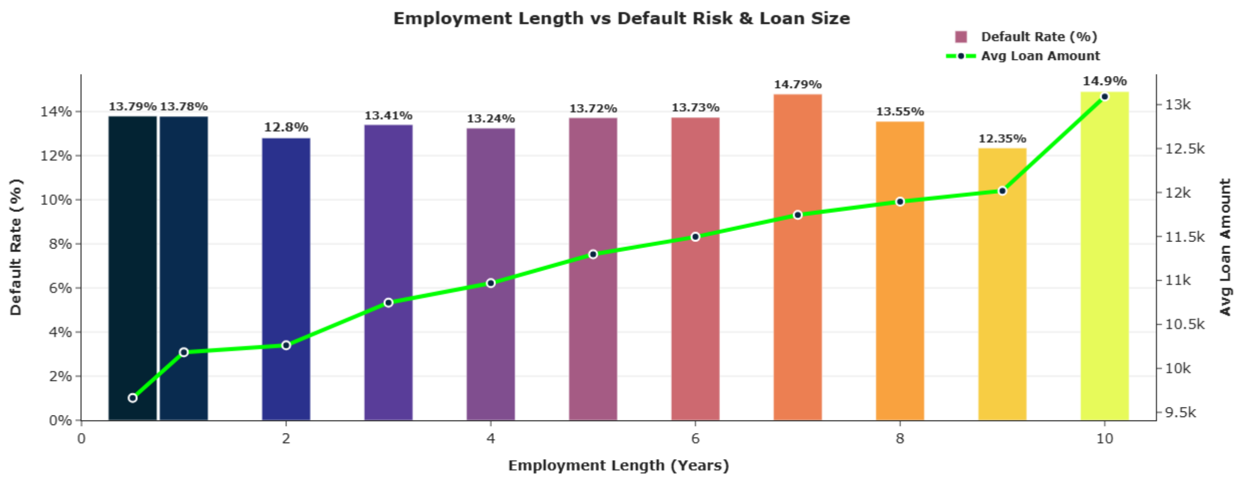



## CUSTOMER & RISK SEGMENTATION

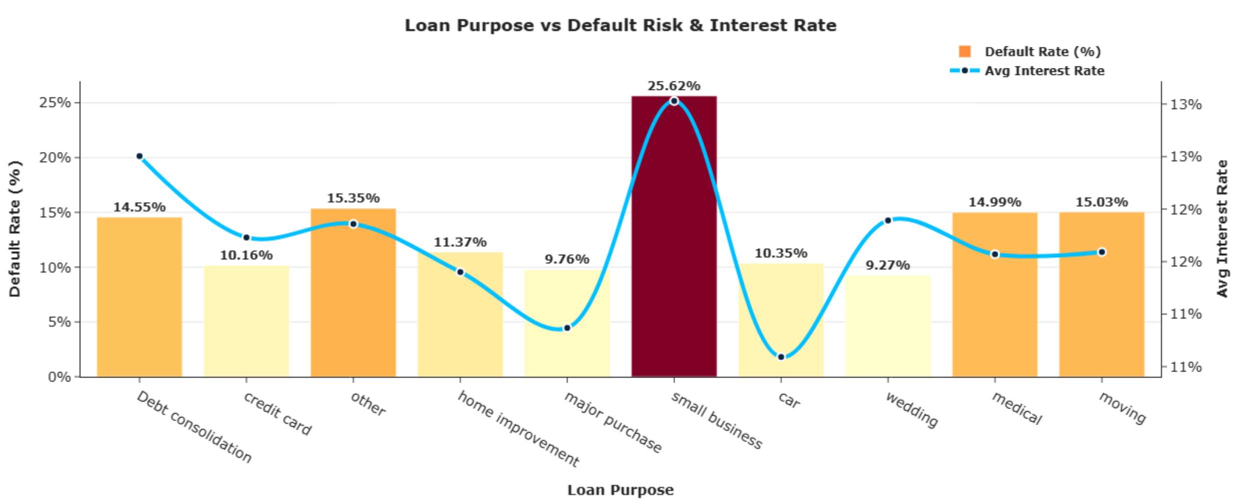


| Income | Default Rate |
|---|---|
| 0 – 30k | 17.3% |
| 30k – 60k | 15.3% |
| 60k – 100k | 12.1% |
| 100k – 200k | 10.4% |
| 200k + | 10.2% |

- Higher the Income lower the default rate.




- Higher the years, higher the **average loan amount.**



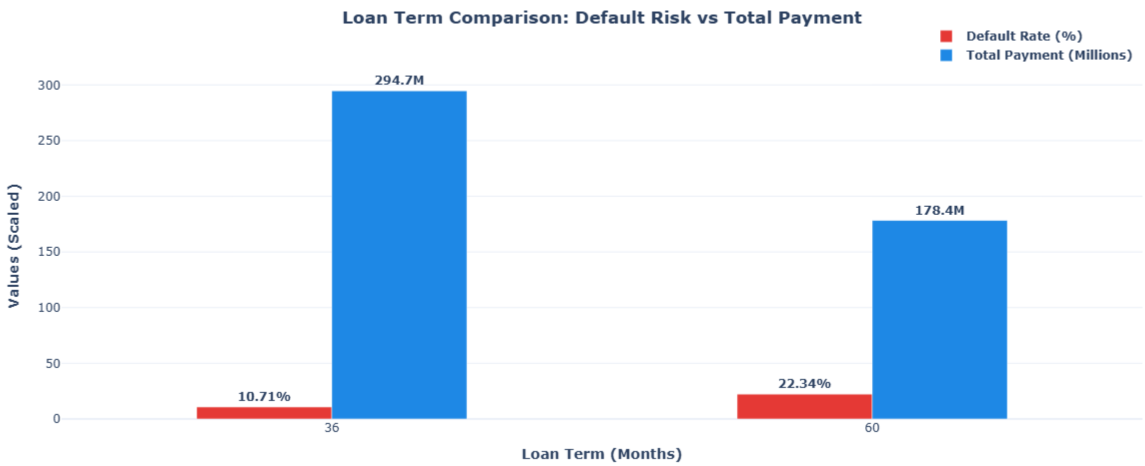

- Highest loan purpose is **starting small business.**
- Lowest loan purpose is for **car.**

## PRICING & REPAYMENT BEHAVIOR

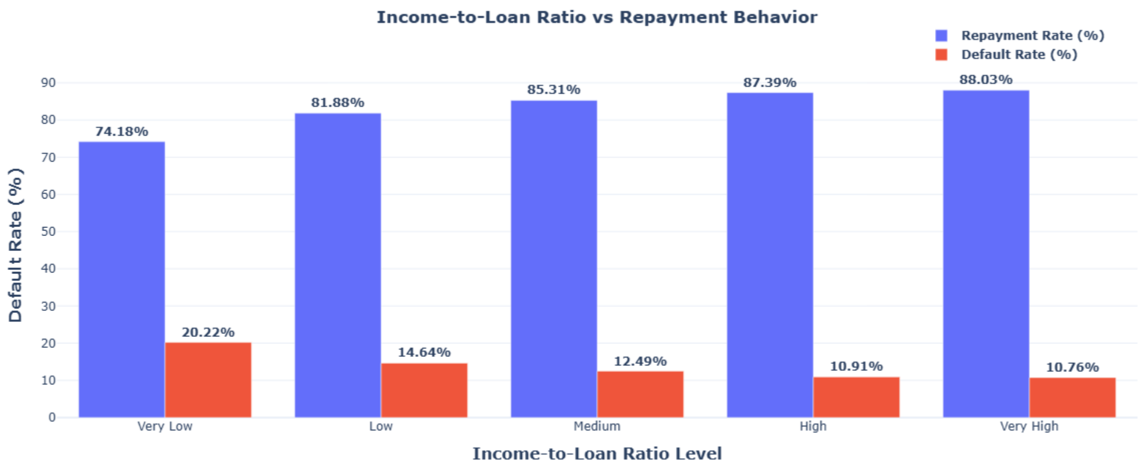
| Grade | Interest Rate |
|---|---|
| A | 7.5% |
| B | 11% |
| C | 13.5% |
| D | 15.7% |
| E | 17.7% |
| F | 19.7% |
| G | 21.4% |

- Higher the grade, higher the **interest rate.**

## CORRELATION BETWEEN INSTALLMENT & DEFAULT: **0.027**

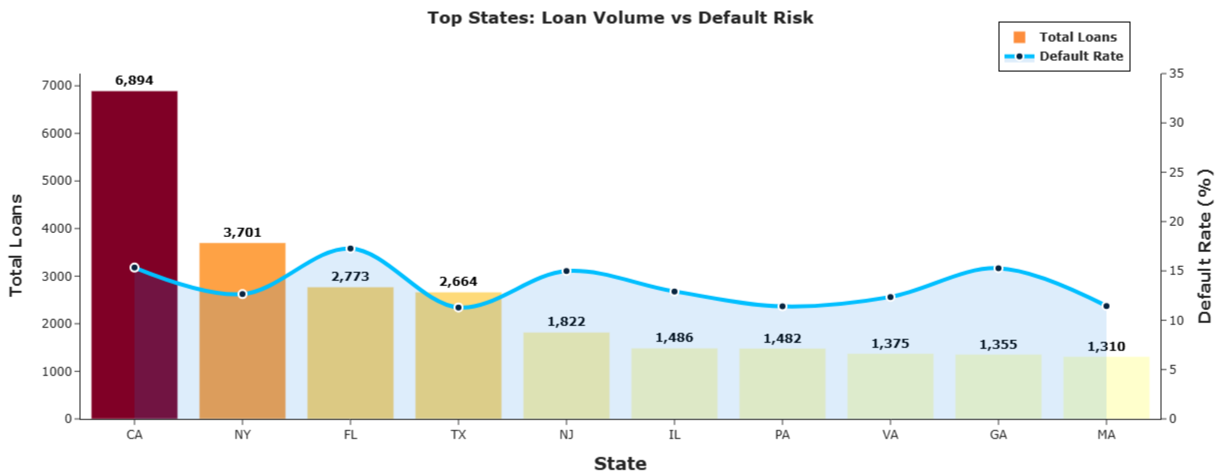



- 36 Months terms has higher Payment than 60 Months term.
- 60 Months term has higher default rate than 36 Months term.



## GEOGRAPHIC & BEHAVIORAL INSIGHTS

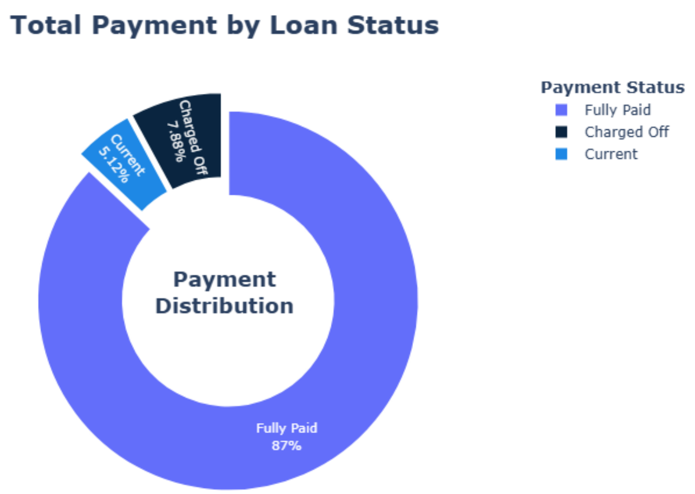
| Countries | Number of Loans | Countries | Number of Loans |
|---|---|---|---|
| CA | 6894 | IL | 1486 |
| NY | 3701 | PA | 1482 |
| FL | 2773 | VA | 1375 |
| TX | 2664 | GA | 1355 |
| NJ | 1822 | MA | 1310 |



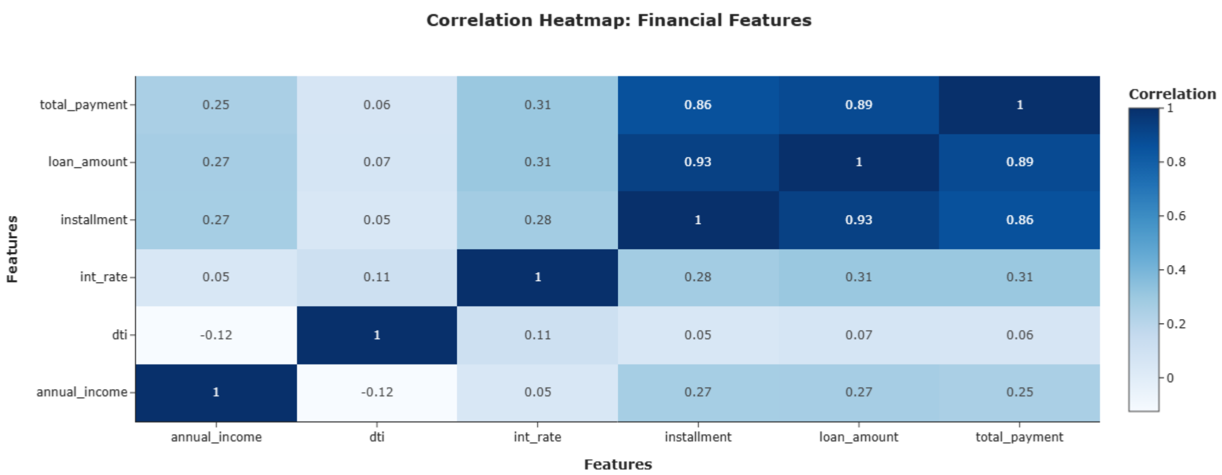

## 1. REVENUE & RISK SIGNALS

- Revenue only from Fully Paid loans: **411,586,256$**

| Loan Status | Total Loans | Total Payment | Payment Share | Total Loss |
|---|---|---|---|---|
| **Charged Off** | 5333 | $37,284,763 | 7.8% | $28,247,462 |
| **Current** | 1098 | $24,199,914 | 5.1% | 0 |
| **Fully Paid** | 32145 | $411,586,256 | 87% | 0 |




## CORRELATION




## BUSINESS RECOMMENDATIONS(SUGGESTIONS)

1. Reduce exposure to high-risk loan grades (**E–G)** as they have significantly higher default rates. Focus more on safer grades like **A** and **B.**
2. Limit long-term loans (**60 months**) since they show much higher default rates compared to **36-month** loans.
3. Strengthen **credit policies** for high-risk purposes such as **small business loans**, which have the highest default rates.

4. Encourage lending to higher-income customers, as default risk decreases significantly with increasing income levels.
5. Apply stricter screening for borrowers with high debt-to-income (DTI) ratios, especially in the **15–25%** range where risk peaks.
6. Promote loans with **higher income-to-loan ratios**, as they show better repayment performance and lower default rates.
7. Improve **monitoring** of high installment loans, as default risk increases with higher installment amounts.
8. Maintain risk-based pricing strategy, as interest rates are effectively aligned with loan risk across grades.
9. Focus **marketing** and **lending** in lower default regions (**e.g., Texas, Pennsylvania**) and review strategies in high-risk states like **Florida**.
10. Minimize losses from charged-off loans by **improving early risk detection** and **strengthening collection strategies.**